House Price Prediction Using Linear Regression

Objective

The objective of this project is to build and evaluate a Linear Regression model using the California Housing dataset. The project demonstrates the complete machine learning workflow, including data loading, exploratory data analysis (EDA), model training, prediction, and evaluation.

Dataset

The California Housing dataset was obtained from Scikit-Learn. The dataset contains housing-related information such as:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

Target Variable:

- Median House Value (MedHouseVal)

The dataset contains 20,640 observations and 8 features.

Exploratory Data Analysis (EDA)

Missing Values

The dataset was checked for missing values using Pandas.

Result:

- No missing values were found.

Correlation Analysis

A correlation heatmap was generated to understand relationships between variables and the target feature.

[INSERT HEATMAP SCREENSHOT HERE]

Observation:

- Median Income (MedInc) has the strongest positive correlation with house prices.
- Geographic variables also influence housing prices.

Distribution Analysis

The distribution of house prices and selected features was visualized to understand data spread and patterns.

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Model Training and Evaluation

Data Splitting

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

A Linear Regression model was trained using Scikit-Learn.

Performance Metrics

Results obtained:

- MAE: 0.533
- RMSE: 0.746
- R^2 Score: 0.576

Actual vs Predicted Values

[INSERT ACTUAL VS PREDICTED PLOT HERE]

Residual Plot

[INSERT RESIDUAL PLOT HERE]

Interpretation:

The model achieved an R^2 score of 0.576, meaning it explains approximately 57.6% of the variation in house prices. The prediction accuracy is reasonable for a baseline Linear Regression model.

Conclusion

A Linear Regression model was successfully built and evaluated using the California Housing dataset.

Key Findings:

- Median Income is the most influential feature.
- The model achieved an R^2 score of 0.576.
- Linear Regression provides a good baseline for house price prediction.

Future Improvements:

- Feature Engineering
- Hyperparameter Tuning
- Random Forest Regression
- Gradient Boosting Models
- XGBoost Regression

These techniques may improve predictive performance beyond the baseline model.